1023 Quiz 2 — Free Response Scoring Guide

Question	Source	Topics (2026 CED)	Key answers
1	2011 AP FRQ #2	2.1, 2.2, 2.6, 2.7: two-way table, conditional probability, independence, segmented bar graph	(a) 48/200 = 0.24 (b) 0.24 ≠ 168/500 = 0.336, not independent (c) both bars 0.176 / 0.488 / 0.336
2	2014 AP FRQ #2	2.3, 2.4, 2.6: multiplication rule without replacement, judging a claim, evaluating a simulation	(a) 3/9 × 2/8 × 1/7 = 1/84 ≈ 0.012 (b) yes, small probability (c) no, dice are independent (with replacement) but the selection is without replacement
3	Original	2.4, 2.5, 2.6, 2.7, 2.3, 2.11: normal probability, complement, mutually exclusive vs. independent, simulation	(a) 0.0548 (b) 0.246 (c) yes ME; not independent (d) 00–04 = late, 5 two-digit numbers

How to score each question (0–4)

Score each part E (essentially correct), P (partially correct), or I (incorrect), then convert to a 0–4 score with the table in that question's guideline. Q1 and Q2 have 3 parts and use the official conversion. Q3 has 4 parts: count each E as 1 point and each P as ½ point; if the total ends in ½, use holistic judgment of the whole response to round up or down.

Apply the AP rule throughout: an incorrect statement showing a serious statistical misunderstanding lowers that part by one level (E → P or P → I), even when the rest of the part is correct. Calculator syntax alone (e.g., normalcdf(36, 1E99, 32, 2.5)) without labeled values does not earn the work component.

Question 1 (2011 AP Statistics FRQ #2) — Solution

(a) Condition on males: 48/200 = 0.24.

(b) P(Party Y | male) = 0.24, but P(Party Y) = 168/500 = 0.336. Because 0.24 ≠ 0.336, the events “is a male” and “is registered for Party Y” are not independent. Students must compute both values, state that they are unequal, and conclude in context.

(c) Independence means each gender has the same distribution as the township overall: W 88/500 = 0.176, X 244/500 = 0.488, Y 168/500 = 0.336. Both bars are divided at 0.176 and 0.664 and labeled W, X, Y.

Watch for: the reverse conditional 48/168 in (a) (scores P); confusing independent with mutually exclusive anywhere in the response (lowers one level).

Official scoring guidelines (2011 SG, pages 6–8)

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Question 2

Intent of Question

The primary goals of this question were to assess students' ability to (1) determine a conditional probability from a table of data; (2) use a table of data to determine whether or not two events are independent; (3) demonstrate an understanding of the concept of independence by constructing a graph that displays independence between two variables.

Solution

Part (a):

Of the 200 male registered voters in Franklin Township, 48 are registered for Party Y. Therefore the conditional probability that a randomly selected voter is registered for Party Y, given that the voter is a male, is $\frac{48}{200} = 0.24$.

Part (b):

No, the events "is a male" and "is registered for Party Y" are not independent. One justification of this conclusion is to note that the conditional probability of the event "is registered for Party Y" given the event "is a male" — which was computed in part (a) — is not equal to the probability of the event "is registered for Party Y," as shown below.

$$P(\text{is registered for Party Y} \mid \text{is a male}) = 0.24$$
$$P(\text{is registered for Party Y}) = \frac{168}{500} = 0.336$$

Because $0.24 \neq 0.336$, the two events are not independent.

Part (c):

The marginal proportions of voters registered for each of the three political parties (without regard to gender) are given below.

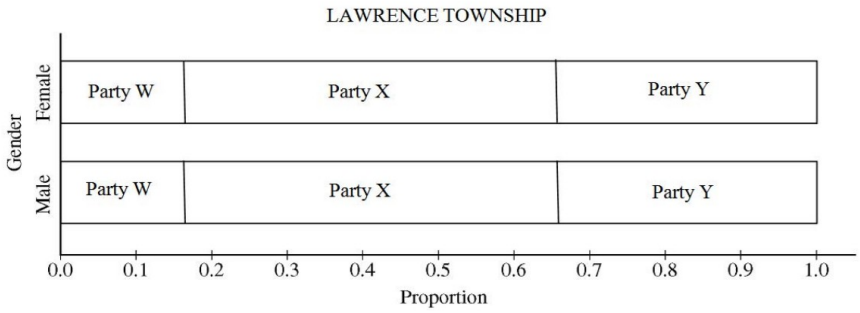
$$\text{Party W: } \frac{88}{500} = 0.176$$
$$\text{Party X: } \frac{244}{500} = 0.488$$
$$\text{Party Y: } \frac{168}{500} = 0.336$$

Because party registration is independent of gender in Lawrence Township, the proportions of males and females registered for each party must be identical to each other and also identical to the marginal proportion of voters registered for that party. Using the order Party W, Party X, and Party Y, the graph for Lawrence Township is displayed below.

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Question 2 (continued)



Scoring

Parts (a), (b) and (c) are scored as essentially correct (E), partially correct (P) or incorrect (I).

Part (a) is scored as follows:

Essentially correct (E) if the response has the correct conditional probability *AND* shows the work.

Partially correct (P) if the response has the correct reverse conditional probability (of being a male given that he is registered for Party Y),
OR
if the response has the correct conditional probability *BUT* does not show work.

Incorrect (I) if the response fails to meet the criteria for E or P.

Part (b) is scored as follows:

Essentially correct (E) if the response identifies two values whose inequality implies a lack of independence between the events *AND* includes the following three components:
1. Correct computations of the two values.
2. An explicit statement of whether the two values are equal or unequal.
3. An appropriate conclusion about the independence of the events.

Partially correct (P) if the response identifies two values whose inequality implies a lack of independence between the events but includes only two of the three components listed above.

Incorrect (I) if the response fails to meet the criteria for E or P.

Part (c) is scored as follows:

Essentially correct (E) if the response shows the same conditional distribution of party registration for both males and females *AND* includes the following two components:
1. Correct proportions for each party.
2. Correct labels (Party W, Party X, Party Y).

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Question 2 (continued)

Partially correct (P) if the response shows the same conditional distribution of party registration for both males and females *AND* includes only one of the two components listed above.

Incorrect (I) if the response fails to meet the criteria for E or P.

Note: For all three parts, an incorrect statement that indicate a serious misunderstanding of statistical concepts, even if unrelated to the rest of the response, lowers the score one level (that is, from E to P, or from P to I). An example of this is a response that indicates confusion between independent events and disjoint events.

4 **Complete Response**

All three parts essentially correct

3 **Substantial Response**

Two parts essentially correct and one part partially correct

2 **Developing Response**

Two parts essentially correct and one part incorrect

OR

One part essentially correct and one or two parts partially correct

OR

Three parts partially correct

1 **Minimal Response**

One part essentially correct and two parts incorrect

OR

Two parts partially correct and one part incorrect

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Question 2 (2014 AP Statistics FRQ #2) — Solution

(a) Without replacement, using the general multiplication rule: $(3/9)(2/8)(1/7) = 1/84 \approx 0.012$.

(b) Yes. If the selection were random, getting all 3 women would happen only about 1.2% of the time. This is small enough to doubt the manager's claim. The response must not say the probability proves the selection was not random.

(c) No. The three dice rolls are independent (each has probability 1/3 of "woman"), which models selection with replacement. The real selection is without replacement, so the probability of choosing a woman changes after each pick. Equivalently: the dice give $(1/3)^3 \approx 0.037$, not 0.012.

Watch for: answering (c) only with "the dice can repeat a person" without stating both the independence of the dice and the dependence of the selections (scores P).

Official scoring guidelines (2014 SG, pages 5–8)

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Question 2

Intent of Question

The primary goals of this question were to assess a student's ability to (1) calculate a probability; (2) assess whether a claim about randomness is questionable in light of a calculated probability; and (3) determine whether a description of a simulation method achieves a correct simulation of a random process.

Solution

Part (a):

The probability that all 3 people selected are women can be calculated using the multiplication rule, as follows:

P(all three selected are women)

$= P(\text{first is a woman}) \times P(\text{second is a woman} \mid \text{first is a woman}) \times P(\text{third is a woman} \mid \text{first two are women})$

$= \frac{3}{9} \times \frac{2}{8} \times \frac{1}{7} \approx 0.012$

Part (b):

The probability calculated in part (a) does provide a reason to doubt the manager's claim that the selections were made at random. The calculation shows that there is only about a 1.2% chance that random selection would have resulted in three women being selected. The probability is small enough that it may cast doubt on the manager's claim that the selections were made at random.

Part (c):

No, the process does not correctly simulate the random selection of three women from a group of nine people of whom six are men and three are women. The random selection of three people among nine is done *without* replacement. However, in the simulation with the dice, the three dice rolls in any given trial are independent of one another, indicating a selection process that is done *with* replacement.

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Question 2 (continued)

Scoring

Parts (a), (b), and (c) were scored as essentially correct (E), partially correct (P), or incorrect (I).

Part (a) is scored as follows:

Essentially correct (E) if the response correctly computes the probability of selecting the three women, and shows how the probability was computed.

Partially correct (P) if the response shows only one of the following:

Gives the correct probability of $\frac{1}{84}$ (0.012 or 0.011 is acceptable) but does not show how it was computed;
OR
Correctly shows how the probability should be computed, but does not carry the computation through correctly;
OR
Correctly computes (showing work) only the numerator, or only the denominator of the correct answer. (For example, $\frac{1}{9}\times\frac{1}{8}\times\frac{1}{7}\approx 0.002$, or $\frac{3}{9}\times\frac{2}{9}\times\frac{1}{9}\approx 0.008$, or $\frac{3}{9}\times\frac{3}{8}\times\frac{3}{7}\approx 0.054$);
OR
Mistakenly assumes independence and calculates (showing work) the binomial probability $\frac{1}{3}\times\frac{1}{3}\times\frac{1}{3}=\frac{1}{27}\approx 0.037$.

Incorrect (I) if the response does not meet the criteria for E or P.

Part (b) is scored as follows:

Essentially correct (E) if the response states that the probability from part (a) is small (or insufficiently small), makes an appropriate decision consistent with the probability being small (or insufficiently small), and does so in the context of this situation.

Partially correct (P) if the response shows only one of the following:

Otherwise satisfies the criteria for an E but does so without any context;
OR
States a significance level and makes a decision in context that is appropriate to the given probability in part (a) and the stated significance level, but does not explicitly compare the probability and the significance level;
OR
Otherwise satisfies the criteria for an E but does not explicitly make a decision about whether there is reason to doubt the manager's claim. (For example: "The probability of selecting the three women from among the nine employees is very small so it is unlikely to occur by chance.")

Incorrect (I) if the response does not meet the criteria for E or P.

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Question 2 (continued)

Notes:

- Each of the following situations is one in which a response that otherwise would be scored as E should be scored as P, and a response that otherwise would be scored as P should be scored as I:
  - The response includes a statement that the small probability *proves* that the manager did not make the selection at random (or any equivalent wording).
  - The response includes a statement that clearly interprets the probability from part (a) to be the probability that the manager selected the three people at random.
- Each of the following situations is one in which the response is scored as I:
  - The decision is inconsistent with the justification (*e.g.*, "The probability is very small, so there is no reason to doubt the manager's claim").
  - The response states or implies that because the selection of three women was *not impossible,* there is *no reason to doubt* the manager's claim.

Part (c) is scored as follows:

Essentially correct (E) if the response answers no *AND* states that the dice outcomes in the proposed simulation are independent *AND* states that the genders of the selected convention attendees are dependent. The table below shows statements that should be considered equivalent to the required statements of independence and dependence.

Independence of dice outcomes	Dependence of genders
• The three dice outcomes are independent. • The probability of rolling a 5 or a 6 is the same on all three dice. • The dice simulation actually simulates sampling with replacement.	• The genders of the three people are dependent (or not independent). • The probability of selecting a woman changes after each selection. • The people are sampled without replacement.

OR

Essentially correct if the response answers no *AND* computes the correct probability that a trial of the simulation will indicate the selection of three women $\left(\frac{1}{3}\times\frac{1}{3}\times\frac{1}{3}\approx 0.037\right)$ *AND* states that the probability is different from the probability found in part (a).

Partially correct (P) if the response correctly answers no and either:

States only that the dice outcomes are independent or states only that the genders of the selected convention attendees are dependent, but not both;

OR

Otherwise meets the criteria for E but has poor communication. An example of poor communication is: "No, because it selects with replacement. It isn't possible for the same person to be selected twice." (There is an apparent shift between the two sentences from describing the simulation to describing the actual selection of people, but that is not made clear.)

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Question 2 (continued)

Incorrect (I) if the response does not meet the criteria for E or P.

Note: Pointing out that a sample of three people is more than 10% of the population of nine people should be considered equivalent to stating that the selection of a woman is not independent among the three people selected to attend the convention.

4 Complete Response

All three parts essentially correct

3 Substantial Response

Two parts essentially correct and one part partially correct

2 Developing Response

Two parts essentially correct and one part incorrect

OR

One part essentially correct and one or two parts partially correct

OR

Three parts partially correct

1 Minimal Response

One part essentially correct and two parts incorrect

OR

Two parts partially correct and one part incorrect

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Question 3 (original) — Solution

(a) Let T = travel time, T ~ N(μ = 32, σ = 2.5). z = (36 − 32)/2.5 = 1.60. P(T > 36) = P(Z > 1.60) = 1 − 0.9452 = 0.0548.

(b) P(not late) = 1 − 0.0548 = 0.9452 for each trip. With independence, P(no late trips in 5) = $(0.9452)^5$ = 0.7544. P(at least one late) = 1 − 0.7544 = 0.2456 ≈ 0.246.

(c)(i) Yes. One trip cannot take both more than 36 minutes and less than 30 minutes, so P(A and B) = 0.

(c)(ii) No. P(B) = P(T < 30) = P(Z < −0.80) = 0.2119 > 0. Then P(A | B) = 0/0.2119 = 0, but P(A) = 0.0548. Because 0 ≠ 0.0548, knowing that B occurred changes the probability of A, so A and B are not independent. (Also accepted: P(A and B) = 0 ≠ P(A)·P(B) = (0.0548)(0.2119) = 0.0116.)

(d) Let the two-digit numbers 00–04 represent a late trip and 05–99 represent an on-time trip. Read 5 two-digit numbers from the table (repeats allowed, because the trips are independent). Record whether at least one of the 5 numbers is between 00 and 04. (Repeating many trials, the proportion of trials with at least one late trip estimates the probability.)

Scoring

Part (a) is essentially correct (E) if the response includes all three components:

1. Identifies the distribution and the boundary, by name or with labeled values: normal, mean 32, SD 2.5, more than 36. A correct z-score with a labeled formula satisfies this.

2. Indicates the correct region (upper tail), e.g., P(T > 36) or a shaded sketch.

3. Gives the correct probability, 0.0548 (accept 0.055).

Part (a) is partially correct (P) if it has two of the three components. Unlabeled calculator syntax counts as missing component 1. Giving 0.9452 (wrong tail) with otherwise correct work is P.

Part (b) is E if the response uses the complement and independence, $1 - (1 - p)^5$, AND computes the correct value using the answer from (a) (follow-through accepted). It is P if the method is correct but the computation is wrong or incomplete (e.g., stops at 0.754), or if the correct value is given with no work. Adding probabilities (5 × 0.0548 = 0.274) or using p^5 is I.

Part (c) is E if both (i) and (ii) are correct with justification: (i) “yes” with the reason that both cannot happen on one trip; (ii) “no” with two correctly computed probabilities that are compared and a conclusion. It is P if only one of (i), (ii) is correct and justified, or if (ii) states the correct conclusion but does not compute and compare probabilities. A response saying A and B are independent because they are mutually exclusive shows a serious misunderstanding: score (c) as I.

Part (d) is E if the response includes both: (1) a digit assignment that gives probability exactly 0.05 for a late trip, and (2) a description of one trial (5 selections, repeats allowed) and what is recorded (whether at least one is late). It is P if only one of the two components is correct, e.g., using single digits with 0 = late (probability 0.10), or not saying what is recorded. Saying to skip repeated numbers is a P (repeats must be allowed for independent days).

Q3 score	**Total of the four parts (E = 1, P = ½, I = 0)**
4 Complete	4 (all four parts E)
3 Substantial	3 (e.g., 3 E and 1 I; 2 E and 2 P)
2 Developing	2 (e.g., 2 E; 1 E and 2 P; 4 P)
1 Minimal	1 (e.g., 1 E; 2 P)
0 No credit	0

A total ending in ½ (for example 3½ from 3 E and 1 P) is rounded up or down by holistic judgment of the whole response, as on the AP exam.